

ASSIGNMENT 2 — CASE STUDY & SEMINAR (MANDATORY)

In-Line Quality Inspection Strategy

Comparative Analysis of Coordinate Measuring Machines (CMM) versus Non-Contact Vision Inspection Systems · Case Study Technical Report

Particulars	Details
Name of the student	
Enrolment / Roll number · Class	
Subject	Automation in Manufacturing · PBL: Case Study Analysis / Seminar
Target COs / SDGs	CO-1, CO-5 · SDG 9, SDG 4
Accompanying file	Assignment2_Seminar_Presentation.pptx (12 slides)

1. Introduction and Problem Context

An automated plant producing **engine cylinder blocks** is losing throughput at the quality-inspection stage. Blocks are currently taken off the line and carried to an offline inspection room for dimensional checking. Two consequences follow. First, inspection has become the bottleneck of the line. Second, and more damaging, the **detection lag** is long: by the time a drifting bore diameter is discovered, a large quantity of work in progress has already been machined to the wrong size.

Plant management must choose between integrating an **in-line or near-line Coordinate Measuring Machine (CMM)** and adopting a **high-speed non-contact vision or laser-scanner system**. This report analyses both, compares them against the parameters that matter for this part, and makes an engineering recommendation.

Basis of the study. The component is a four-cylinder cast-iron cylinder block, approximately 600 × 400 × 300 mm and 38 kg. The line runs to a **90 s takt** — 40 blocks per hour, 800 per day on two shifts. The critical-to-quality features are cylinder bore diameter, roundness and cylindricity; bore-to-bore true position; deck-face flatness; main-bearing bore alignment; and the presence and depth of the oil-gallery drillings and threads.

A note on the standards cited — the brief cites **IS 2063 (Parts 1 and 7)** and **IS 15250**. IS 15250 is the Indian Standard series for the acceptance and reverification of coordinate measuring machines and corresponds to **ISO 10360**; the technical content below is anchored on ISO 10360 and ISO 230, whose provisions are described directly. The exact part numbers and current titles should be confirmed against the live BIS catalogue before the report is submitted, since the series are periodically revised.

2. Tactile CMM Architectural Analysis

2.1 Structural configurations

Configura tion	Construction	Strengths	Weaknesses	Best suited to
Moving bridge	Bridge spans a granite table and travels in Y; a carriage traverses X on the bridge; the quill moves in Z. The part stays still.	Best accuracy-to-cost of all types; stiff closed structure; part mass does not affect the moving dynamics; the most widely supported type	Bridge "walking" (yaw) if driven from one side — countered by centre-of-gravity or dual drive; access limited to the top	Medium parts needing high accuracy — the natural choice for a cylinder block
Fixed bridge	The bridge is rigidly fixed to the base and the TABLE moves in Y.	Highest accuracy of the ball-and-socket types, because the measuring structure never moves	Moving table limits part weight severely; roughly double the floor area for a given volume; expensive	Small, light, very high precision parts — gauge rooms and calibration laboratories
Cantilever	The probe is carried on an arm cantilevered from one side of the table.	Open on three sides, so loading is easy for an operator or a robot; small footprint; fast	The cantilever deflects as it extends, giving a varying Abbe error; lowest stiffness; limited reach	Small, light parts with frequent loading — turned and milled components
Gantry	A self-supporting frame stands on its own floor rails, independent of the part, which sits on the floor or a heavy base.	Very large working volumes; the part can be craned in; the operator can walk around it	Lower accuracy per unit length than a bridge; large and expensive; needs a good foundation	Very large and heavy workpieces — machine beds, large castings, body-in-white
Horizontal arm	A horizontal quill moves in Y from a column that travels in X, the arm rising and falling in Z.	Excellent access to vertical faces; large volume at modest cost; often used in opposed pairs	Arm droop varies with extension, so it is the least accurate of the four	Sheet-metal assemblies and car bodies, where access matters more than micrometres

Recommendation — heavy versus light machine parts

Part class	Recommended configuration	Reasoning
Heavy parts — cylinder blocks, gearbox housings, large castings (20–200 kg)	Moving-bridge CMM on a granite table (gantry only if the part exceeds roughly 1 500 kg or 3 m)	The part remains stationary, so its mass never enters the servo dynamics and cannot excite the structure. The closed bridge loop gives the stiffness needed for scanning. A fixed bridge is ruled out because a 38 kg block on a moving table would degrade accuracy and limit acceleration.
Light parts — turned shafts, small milled components, pressings (< 5 kg)	Cantilever CMM , or a small moving-bridge machine where tighter accuracy is required	Open access on three sides suits frequent manual or robot loading and shortens the load cycle. The lower stiffness of the cantilever is acceptable because the measuring volume, and therefore the arm extension, is small.

2.2 Drive control systems

Type	How it works	Throughput	Repeatability	Verdict for this plant
Manual (free-floating)	Air bearings released; the operator pushes the carriages by hand and triggers each point	Very low	Poor — depends entirely on approach speed and operator skill	Rejected — unsuitable for production, cannot scan

Type	How it works	Throughput	Repeatability	Verdict for this plant
Motorised / joystick-driven	Servo axes driven through a joystick at a controlled speed; the operator still decides the path	Low to moderate	Better, because the probe approach speed is consistent	Rejected for routine work; useful only for one-off layout and for teaching a program
Direct Computer Control (DCC)	A CNC-type part program drives the probe automatically; alignment, measurement, evaluation and reporting are all automatic	High	Excellent — the path and approach speed are identical every time	Selected. DCC is mandatory for a production inspection cell: it alone allows unattended operation, line triggering, automatic part alignment and continuous scanning

2.3 Probe systems

	Touch-trigger probe	Continuous contact scanning probe
Principle	A kinematic seat or strain gauge generates a trigger pulse on contact; the controller latches the scale readings at that instant	The stylus stays in contact and its deflection is measured continuously in three axes while the head sweeps the surface
Data rate	3–5 points per second in production	Several hundred to several thousand points per second
What it measures well	Sizes, positions, distances — features defined by a few points	Size and form — roundness, cylindricity, straightness, flatness, profile of a surface
Principal error source	Pre-travel variation ("lobing") — the trigger force varies with the direction of approach. Strain-gauge designs reduce it substantially	Dynamic error at speed; needs a stiffer machine and careful control of scanning speed and stylus mass
Relative cost	Low	High (probe, controller option and a machine stiff enough to use it)
Suitability here	Adequate for hole positions and thread presence	Required. Bore roundness and cylindricity on a cylinder block cannot be certified from a handful of discrete points

Probe selection — a **continuous contact scanning head with a touch-trigger capability** is specified. The cylinder bores demand form evaluation, and collecting the several hundred points per bore that a credible cylindricity assessment needs would take a touch-trigger probe many minutes per bore.

2.4 Static accuracy, laser-interferometric calibration and error compensation

A CMM is a measuring instrument and must itself be calibrated and periodically reverified. Its geometry is described by **21 rigid-body error components**:

Error group	Count	Description
Linear positioning (scale) error	3	One per axis — the difference between commanded and actual travel along that axis
Straightness errors	6	Two per axis — deviation of the carriage path from a straight line in the two directions perpendicular to travel
Angular errors	9	Three per axis — pitch, yaw and roll of the carriage as it travels
Squareness errors	3	One per axis pair — XY, YZ and ZX departure from 90°
Total	21	Together these define the volumetric error at every point in the measuring volume

Measurement of these errors

Laser interferometry is the primary instrument. A displacement-measuring interferometer gives the **linear positioning error** of each axis directly against the wavelength of a stabilised laser, with sub-micrometre uncertainty; with angular optics the same instrument measures **pitch and yaw**, and with Wollaston straightness optics the two **straightness** components. **Roll** is measured separately with a precision electronic level or an autocollimator, and **squareness** with a calibrated granite square or by measuring a **ball plate or ball bar** in several orientations. Throughout, the interferometer must be fed air temperature, pressure and humidity so that the refractive index of air is corrected, and material temperature so that scale and artefact are reduced to 20 °C.

Positional error compensation

The measured 21 components are loaded into the controller as a **Computer Aided Accuracy (CAA) volumetric compensation map**. Every measured point is then corrected in real time before it is reported. Volumetric accuracy typically improves by a factor of three to ten; CAA is the reason a modern production CMM can hold a few micrometres over a cubic metre of measuring volume.

Acceptance and reverification tests

Test	Standard	What is done	What is reported
Length measurement error	ISO 10360-2 (IS 15250 series)	A calibrated step gauge or gauge-block stack is measured at five positions and in seven orientations within the volume, three times each	E_o,MPE stated as A + L/K μm, e.g. 1.9 + L/300
Repeatability	ISO 10360-2	Repeated measurement of the same calibrated length	R_o,MPL — the range of repeated results
Probing error (form)	ISO 10360-5	25 points are taken over a hemisphere of a calibrated test sphere and a best-fit sphere computed	P_FTU,MPE — the range of the radial deviations
Scanning probing error	ISO 10360-4	Four prescribed lines are scanned on the test sphere within a stated time limit	T_HP,MPE — the range of radial deviation, and the time taken
Geometric accuracy of the structure	ISO 230 series (and IS 2063 as cited in the brief)	The 21 components are measured as described above	Individual error curves, which become the CAA table

Practical point — the specified accuracy is only valid inside the stated thermal window. A CMM quoted at $1.9 + L/300 \mu\text{m}$ over $20 \pm 1^\circ\text{C}$ will not deliver that on an open shop floor at 28°C . This single fact is what forces the CMM in this plant to be **near-line in a climate-controlled room** rather than truly in-line.

3. Non-Contact Inspection and Machine Vision Evaluation

3.1 Three-dimensional laser scanners

A laser line is projected onto the surface and viewed by a camera set at a known angle. The displacement of the imaged line gives the stand-off distance by **triangulation**; sweeping the line across the part builds a dense point cloud. The sensor may be carried on a CMM quill, on a robot arm, or fixed above a conveyor.

- **Speed and output** — 10 000 to 600 000 points per second, giving a full-field point cloud in two to five minutes against 25 minutes for a tactile CMM. The cloud can be compared directly against the CAD model to produce a colour deviation map, which a tactile CMM cannot practically produce.
- **Accuracy and traceability** — typically $20\text{--}50 \mu\text{m}$ for a single point, improving to $10\text{--}30 \mu\text{m}$ for a feature fitted through many points. Acceptance is to **VDI/VDE 2634 Parts 2 and 3** rather than ISO 10360, which is the usual difficulty when a customer demands traceability.
- **Weakness** — strongly affected by surface finish. A freshly machined, shiny cast-iron deck face reflects specularly and can saturate the camera or drop out altogether; dark and translucent surfaces scatter. Matting spray solves it in a laboratory but is not acceptable in-line.

Structured-light (fringe-projection) sensors work on the same triangulation principle but project a full pattern rather than a line, capturing an entire field in a few seconds at $10\text{--}30 \mu\text{m}$. They are a credible alternative where the part can be brought to rest.

3.2 CCD / CMOS machine vision systems

A camera with a **telecentric lens** views the feature, usually against collimated back-lighting so that the feature appears in silhouette. Sub-pixel edge-detection algorithms locate the edge to a fraction of a pixel.

Aspect	Detail
Resolution model	For a 50 mm field of view on a 5 MP sensor (2 448 px across), one pixel $\approx 20 \mu\text{m}$. Sub-pixel interpolation at 1/10 pixel gives $\approx 2 \mu\text{m}$ of *repeatability*; realistic *accuracy* after calibration, lighting drift and thermal effects is $10\text{--}30 \mu\text{m}$.
Why telecentric optics	A telecentric lens has constant magnification over its depth of field, so a part that sits slightly higher or lower does not appear larger or smaller. Without it, dimensional vision measurement is not credible.
CCD versus CMOS	CCD historically offered better uniformity and lower noise. Modern CMOS sensors have overtaken CCD on frame rate, cost, dynamic range and global-shutter availability, and now dominate industrial vision. CMOS with a global shutter is specified , because a rolling shutter distorts a part that is still moving.
Speed	Image capture and analysis in well under one second per view; a complete multi-camera station comfortably fits a 90 s takt.

Aspect	Detail
Limitations	Two-dimensional unless several cameras or a stereo pair are used; measures only what it can see, so blind bores and internal galleries are out of reach; absolutely dependent on controlled, stable lighting.

3.3 Comparison across the key parameters

Parameter	Tactile CMM (DCC + scanning)	3-D laser scanner	CCD / CMOS vision
Measuring range	500–2 000 mm per axis typical; gantry types to several metres	Stand-off 50–500 mm; large volumes covered by moving the sensor on a robot	Field of view 10–300 mm per camera; larger coverage needs more cameras
Speed / throughput	Touch-trigger 3–5 pts/s; scanning several hundred pts/s. ≈ 25 min per block	10 000–600 000 pts/s. ≈ 2–5 min per block	Sub-second per view. ≈ 45 s for the full CTQ set
Accuracy	Best — E_0 , MPE $\approx 1.9 + L/300$ μm (shop-floor grade)	20–50 μm per point, 10–30 μm fitted	10–30 μm in 2-D with telecentric optics
Repeatability	0.5–2 μm	5–20 μm	2–10 μm
Surface reflectivity sensitivity	None — contact is mechanical, finish is irrelevant	High — specular, dark or translucent surfaces cause noise and dropout; matting spray impractical in-line	Moderate — back-lit silhouette is robust; front-lit measurement on a shiny machined face is not
Coolant mist, dust, swarf	Part must be clean and dry; swarf on a datum gives a wrong answer; the stylus itself is unaffected	Mist and airborne dust scatter the beam, producing noise and dropout	Degrades edge contrast; requires a sealed, air-purged enclosure and an air knife
Vibration	Most sensitive — needs an isolated base, especially for scanning	Moderately tolerant because exposure times are short	Most tolerant — strobed illumination freezes motion
Temperature	Needs $20 \pm 1-2$ °C or full compensation	Less sensitive itself, but the part still expands	Same — the part still expands
Blind and internal features	Yes — bent and star styli reach into bores, undercuts and galleries	Line of sight only	Line of sight only
GD&T capability	Full — datum systems, true position, cylindricity, runout, profile	Form and profile strong; datum-referenced GD&T possible but harder to defend	Limited — essentially 2-D sizes and positions
Traceability standard	ISO 10360 / IS 15250	VDI/VDE 2634	No single acceptance standard; relies on gauge R&R
Indicative capital cost	₹ 35–90 lakh	₹ 25–60 lakh	₹ 8–25 lakh per station
In-line capability	Near-line at best for a full block	Feasible in-line on a robot	Excellent — true in-line

Cost figures — the capital costs above are **indicative order-of-magnitude ranges** intended to show the relative scale of the three options. They must be replaced with quotations before any investment decision is taken.

4. Economic and Technical Recommendation Matrix

Each option is scored from 1 (poor) to 5 (excellent) against weighted criteria. The weights reflect this plant's problem: the bottleneck is throughput, but the reason for inspecting at all is to certify a tolerance scheme.

Criterion	Weight	Tactile CMM only	3-D vision / laser only	Hybrid (both)
Inspection cycle time and throughput	25 %	1	5	5
Accuracy and resolution on the CTQ features	20 %	5	3	5
GD&T and datum-referenced capability	15 %	5	2	5
Suitability for automated in-line integration	15 %	2	5	5
Robustness to coolant mist, dust and vibration	10 %	2	3	3
Capital and operating cost	10 %	2	4	3
Software integration, SPC and MES data	5 %	4	4	5
Weighted total (out of 5)	100 %	2.90	3.80	4.60
Rank		3	2	1

4.1 Throughput analysis — why a CMM alone cannot be in-line

Line takt time	=	90 s per block
Full GD&T inspection on the CMM	=	25 min = 1 500 s
CMMs needed for 100 % inspection	=	1 500 / 90 = 16.7 -> 17 machines
One CMM working 20 h per day : 20 x 60 / 25	=	48 blocks inspected
Line output 40 /h x 20 h	=	800 blocks produced
Achievable sampling rate = 48 / 800	=	6 % (1 block in 17)
In-line vision + laser station	=	45 s < 90 s takt
Coverage	=	100 % of blocks

The arithmetic settles the architectural question. Seventeen CMMs is not an engineering proposal, so **a CMM can only ever sample**. A sampling scheme of one block in seventeen means that a process drift is, on average, discovered only after sixteen further blocks have been machined — which is precisely the failure the plant is trying to eliminate.

4.2 Indicative economic model

The value of 100 % in-line gauging lies almost entirely in the **reduction of detection lag**, not in labour saving.

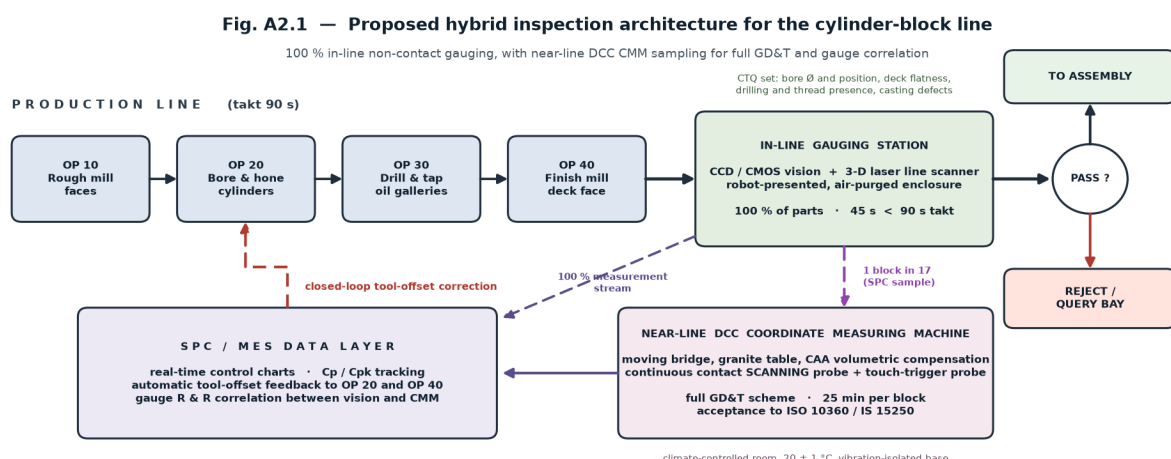
Quantity	Symbol	Illustrative value
Annual production	N	200 000 blocks
Value of a scrapped block (material + machining to that point)	V	₹ 4 500

Quantity	Symbol	Illustrative value
Present scrap and rework rate	S ₁	2.0 %
Expected rate with 100 % in-line gauging and closed-loop offset feedback	S ₂	0.5 %
Annual saving = N × V × (S ₁ – S ₂)		200 000 × 4 500 × 0.015 = ₹ 1.35 crore per year
Capital: in-line vision + laser station		≈ ₹ 45 lakh
Capital: near-line DCC scanning CMM with climate-controlled room		≈ ₹ 85 lakh
Total capital		≈ ₹ 1.30 crore
Simple payback		≈ 1.0 year

These figures are illustrative — they demonstrate the *structure* of the business case, which is that the saving is driven by the scrap-rate improvement and is roughly an order of magnitude larger than the difference in capital cost between the options. The scrap rates and the block value must be replaced with the plant’s own measured data before the case is presented for approval.

5. Engineering Recommendation

Recommendation — adopt a **hybrid architecture**: a **100 % in-line non-contact gauging station** (CMOS vision with telecentric optics, plus a 3-D laser line sensor) immediately after the final machining operation, backed by a **near-line DCC moving-bridge CMM with a continuous scanning probe** in a climate-controlled room, sampling one block in seventeen. Do **not** choose one technology to the exclusion of the other.



Why the hybrid, and not either technology on its own

- A full-GD&T CMM inspection takes 25 min against a 90 s takt — 17 machines would be needed for 100 % coverage, so a CMM alone can only ever sample.
- Non-contact gauging alone cannot reach blind bores, and cannot certify datum-referenced true position and cylindricity to a traceable standard.
- Together, vision catches every part and detects a process drift within one part, while the CMM certifies the measurement system and the full tolerance scheme.

Fig. A2.1 — The recommended hybrid inspection architecture, showing the 100 % in-line gauging station, the near-line sampling CMM, and the closed-loop feedback of tool offsets to the machining stations.

5.1 Division of responsibility

The **in-line station** gauges 100 % of blocks in 45 s, inside the takt: bore diameter and position at defined planes, deck-face flatness, presence and depth of drillings, thread presence and visible casting defects. It catches every part, detects a process drift within one part, and feeds tool offsets back automatically. The **near-line CMM** takes one block in seventeen, plus first-off and last-off of every batch, and spends 25 min off the critical path certifying the complete GD&T scheme — cylindricity, main-bearing bore alignment, datum-referenced true position and surface profile. It is also the reference instrument against which the in-line gauge is correlated.

5.2 Why each technology alone was rejected

1. **A CMM alone cannot meet the takt.** 1 500 s of inspection against a 90 s takt would need seventeen machines. Sampling at one in seventeen leaves the detection lag that caused the problem in the first place.
2. **Non-contact gauging alone cannot certify the part.** It is line-of-sight, so blind bores, oil galleries and main-bearing bores are unreachable, and it has no equivalent of ISO 10360 to underwrite datum-referenced true position and cylindricity to a customer.
3. **Together they are complementary, not redundant.** The in-line station provides coverage and speed; the CMM provides accuracy and traceability, and — importantly — acts as the **reference instrument against which the in-line gauge is periodically correlated by gauge R&R study**. Without that correlation the in-line readings would have no defensible link to the national standard.

5.3 Two conditions for success

First, **correlate the gauges formally**: run a gauge R & R study between the in-line station and the CMM on the same blocks, repeat it quarterly, and set the in-line control limits inside the drawing tolerance by the measured uncertainty of the in-line gauge, so that a marginal part is never passed by the faster instrument. Second, **protect the optics** with a sealed, filtered, positive-pressure enclosure and an air knife that clears coolant before imaging — most in-line vision installations that fail do so for this reason rather than for want of resolution.

6. Conclusion

The choice presented to the plant — CMM **or** vision — is a false alternative. The throughput arithmetic rules out a tactile CMM as a 100 % in-line instrument, and the physics of line-of-sight measurement rules out non-contact gauging as a complete substitute for one. The weighted trade-off matrix scores the hybrid architecture at **4.60 out of 5** against 3.80 for non-contact alone and 2.90 for a CMM alone.

The recommended architecture gauges every block in-line within the 90 s takt, certifies the full tolerance scheme on a sampled basis against ISO 10360, and closes the loop back to the machining stations so that a drift is corrected within a single part rather than within seventeen. On illustrative figures the capital of approximately ₹1.30 crore is recovered in about one year from the scrap reduction alone.